

MD KHALEQUZZAMAN CHOWDHURY SAYEM

Researcher, Vision & Learning Lab (UNIST), South Korea

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Research Summary

Researcher in **3D vision and vision-language models**, focusing on geometry-grounded perception and reasoning for embodied systems. My work spans real-time two-hand manipulation modeling (**AAAI 2025**) and the development of **HandVQA (CVPR 2026)**, a **1.6M-scale** benchmark diagnosing spatial reasoning failures in foundation VLMs. Across these projects, I integrate explicit **3D geometric supervision** with multimodal Transformers to improve generalization in manipulation and language-grounded tasks.

Research Interests

Physical AI; Vision-Language-Action (VLA) Models; Geometry-Grounded World Models; Multimodal Representation Learning; 3D Human-Object Interaction; Embodied Intelligence.

Education

M.S. in Computer Science and Engineering

Ulsan National Institute of Science & Technology (UNIST)

Sep 2023 – Aug 2025

Ulsan, Korea

– GPA: 3.93/4.3 (96.30/100)

– **Master's Thesis (Accepted at CVPR 2026):** *HandVQA: Diagnosing and Improving Fine-Grained Spatial Reasoning about Hands in Vision-Language Models.*

– Advisors: Prof. Seungryul Baek (Supervisor), Prof. Binod Bhattacharai (Co-supervisor)

B.S. in Industrial Engineering

Ulsan National Institute of Science & Technology (UNIST)

Sep 2019 – Aug 2023

Ulsan, Korea

Conference Publications (* equal contribution)

- **[CVPR 2026] MD K. C. Sayem***, M. T. Chowdhury*, Y. Y. Tiruneh, M. A. Khan, M. S. Ali, B. Bhattacharai, S. Baek. *HandVQA: Diagnosing and Improving Fine-Grained Spatial Reasoning about Hands in Vision-Language Models.*
Project Page: kcsayem.github.io/handvqa/
 - Proposed **HandVQA**, a **1.6M-scale** benchmark derived deterministically from **3D hand pose annotations** across FreiHAND, InterHand2.6M, and FPFA.
 - Diagnosed systematic failures of large VLMs (e.g., LLaVA, Qwen2.5VL) in fine-grained spatial reasoning, revealing depth ambiguity, left/right confusion, and relational bias.
 - Demonstrated that injecting **structured 3D supervision** improves zero-shot transfer across tasks, boosting gesture recognition by **+10.33%** and hand-object interaction recognition by **+2.63%**.
 - Provides a scalable pathway for injecting explicit **3D geometric priors** into text-first foundation models.
- **[AAAI 2025] E. Ismayilzada***, **MD K. C. Sayem***, Y. Y. Tiruneh, M. T. Chowdhury, M. Boboev, S. Baek. *QORT-Former: Query-optimized Real-time Transformer for Understanding Two Hands Manipulating Objects.*
Project Page: kcsayem.github.io/QORT-Former/
 - Designed the core query optimization method and integrated structured contact-map features into the Transformer architecture.
 - Proposed a **query division strategy** that separates queries into left-hand, right-hand, and object groups via a dedicated query proposal module, reducing ambiguity between symmetric hand features.
 - Reduced Transformer complexity from **300 queries / 6 decoders** to **108 queries / 1 decoder**, achieving **53.5 FPS** real-time inference while improving state-of-the-art accuracy by **+27.2%** (H2O) and **+10.4%** (FPFA).

Selected Research Projects

Worker Safety System for Industrial Environments Using Multi-Task YOLOv11

Feb 2025 - Apr 2025

Hyundai Motors & Vision and Learning Lab (UNIST) — [GitHub](#)

- Designed and implemented a unified multi-task perception system performing detection, instance segmentation, and human pose estimation.
- Integrated YOLOv11 detection, segmentation, and pose heads into a single pipeline, reducing inference latency by **35%**.

- Implemented text-conditioned editing for dynamic 3D scenes by integrating *InstructPix2Pix* with 4D Gaussian representations.
- Achieved globally consistent edits with fast training (30 min/scene) and high rendering speed (up to 82 FPS).

Research Experience

Vision & Learning Lab, UNIST [↗](#)

Jan 2023 – Present

Researcher

Ulsan, Korea

Graduate Research Assistant (Sep 2023 – Aug 2025); Undergraduate Research Assistant (Jan 2023 – Aug 2023)

- Designed geometry-aware Transformer architectures for real-time 3D hand-object interaction modeling, integrating structured contact and pose representations (AAAI 2025).
- Led large-scale experimentation across FreiHAND, InterHand2.6M, and FPHA to analyze cross-view generalization, occlusion robustness, and representation bias in multimodal systems.
- Developed deterministic data generation and diagnostic evaluation pipelines for fine-grained spatial reasoning in foundation VLMs (CVPR 2026).

Machine Learning & Vision Lab, UNIST [↗](#)

Sep 2021 – Feb 2022

Undergraduate Research Assistant

Ulsan, Korea

- Implemented data augmentation and training improvements for 2D human pose estimation, improving robustness on COCO-style benchmarks.

Technical Skills

- **Core:** PyTorch, CUDA, Distributed/Accelerated Training, Linux, Git, LaTeX
- **Multimodal / Foundation Models:** HuggingFace Transformers, LoRA fine-tuning, VLM benchmarking (e.g., LLaVA, Qwen-VL)
- **3D Vision:** Pose estimation, multi-view geometry, human/hand modeling, NeRF / Gaussian Splatting (projects)
- **Data & Tooling:** NumPy, Pandas, OpenCV, experiment scripting and evaluation pipelines

Teaching Experience

- **Teaching Assistant:** Introduction to AI Programming in Python (Fall 23, Spring 24, Fall 24) — Instructor: Prof. Michael Jon Burrell
- **Teaching Assistant:** Discrete Mathematics (Fall 24) — Instructor: Prof. Michael Jon Burrell
- **Teaching Assistant:** Advanced Computer Vision (Spring 24) — Instructor: Prof. Seungryul Baek

Leadership

PAIS — Partnership for AI and Society

Jan 2021 – Aug 2025

President

UNIST

- Founded and led an AI-safety discussion group with faculty mentorship, organizing talks and forums on data privacy and responsible AI at UNIST.

Honors & Awards

- **Graduate School Scholarship (Korean Government):** full tuition + stipend for M.S.
- **UNIST Dream Scholarship (UNIST):** full tuition + stipend for B.S.
- **AI Challengers Program (UNIST Innovative Education Center):** secured **\$5K** project funding (bandit algorithm parallelization).

REFERENCES

Professor Seungryul Baek

Supervisor

Masters

- **Affiliation:** Associate Professor, Artificial Intelligence Graduate School (AIGS), UNIST, South Korea
- **Affiliation:** Associate Professor, Department of Computer Science and Engineering, UNIST, South Korea
- **Email:** srbaek@unist.ac.kr

Professor Binod Bhattarai

Co-supervisor

Masters

- **Affiliation:** Lecturer, University of Aberdeen, Aberdeen, UK
- **Affiliation:** Honorary Lecturer, University College London, UK
- **Email:** b.bhattarai@ucl.ac.uk